

Assignment–8

OP-AMP APPLICATIONS: INTEGRATOR, DIFFERENTIATOR AND HALF-WAVE PRECISION RECTIFIER

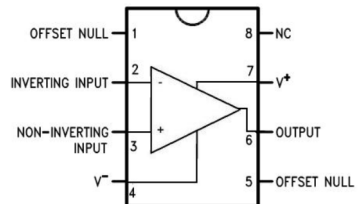
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Objective

- To design and implement Integrator, Differentiator and Half-wave Precision Rectifier using Op-Amp.
- To apply different input signals (sine, square, triangular).
- To observe and analyze output waveforms.

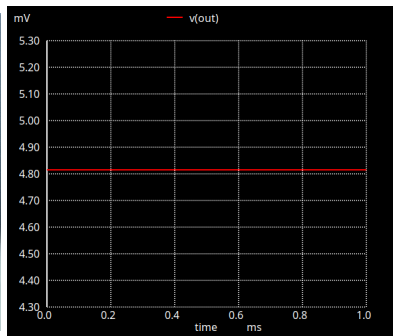
Components Required

- Op-Amp (LM741)
- Resistors
- Capacitors
- Diode
- Function Generator
- Oscilloscope (CRO)
- Dual Power Supply ($\pm 15V$)



Input Offset Voltage Measurement

Before performing the experiments, the input offset voltage of the op-amp was measured. The input terminals were grounded, and the output was observed using the CRO.



Calculation

In this experiment, no external amplification network was used for offset measurement. Hence, the effective closed-loop gain is:

$$A_v \approx 1$$

From CRO observation:

- Measured output offset voltage: $V_{out,offset} \approx 4.816 \text{ mV}$

Therefore, the input offset voltage is:

$$V_{os} = \frac{V_{out,offset}}{A_v}$$

$$V_{os} = \frac{4.816}{1} = 4.816 \text{ mV}$$

$$V_{os} \approx 4.8 \text{ mV}$$

Inference: Since no gain stage is used, the measured output offset directly represents the input offset voltage. The observed value is small and lies within the acceptable range for the LM741 op-amp.

Integrator Circuit

GIVEN DATA

- Resistance, $R = 15 \text{ k}\Omega = 15000 \Omega$
- Capacitance, $C = 0.1 \mu\text{F} = 1 \times 10^{-7} \text{ F}$

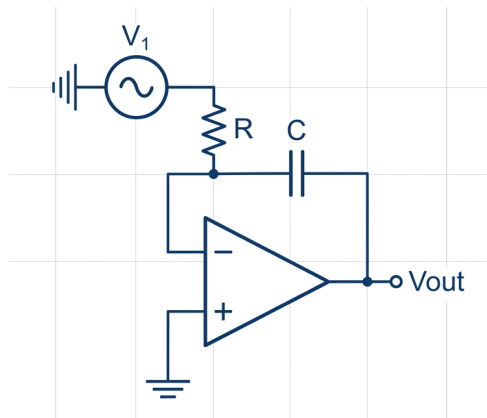
- Frequency, $f = 220 \text{ Hz}$
- Input voltage, $V_{in(pp)} = 4V \Rightarrow V_{in(peak)} = 2V$

THEORY

An op-amp integrator is a circuit in which the output voltage is proportional to the time integral of the input voltage.

$$V_{out}(t) = -\frac{1}{RC} \int V_{in}(t) dt$$

CIRCUIT DIAGRAM

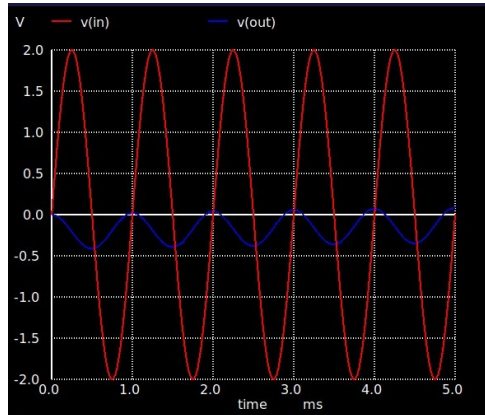
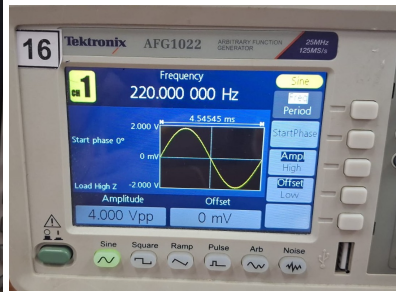
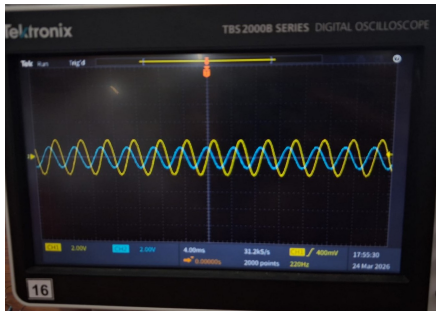


OBSERVATION

Input and output waveforms were observed using CRO.

0.1 For sinusoidal input

- Output is sinusoidal with 90° phase shift.
- Amplitude is reduced as predicted.



For a sinusoidal input $V_{in}(t) = V_m \sin(\omega t)$, the integrator output is:

$$V_{out}(t) = -\frac{1}{RC} \int V_m \sin(\omega t) dt$$

Hence, the output amplitude is inversely proportional to frequency:

$$|V_{out}| = \frac{V_m}{\omega RC}$$

IDEAL GAIN OF INTEGRATOR

$$|A| = \frac{1}{\omega RC}$$

$$\omega = 2\pi f = 2\pi \times 220 = 1382.3 \text{ rad/s}$$

$$RC = 15000 \times 1 \times 10^{-7} = 0.0015$$

$$\omega RC = 1382.3 \times 0.0015 = 2.073$$

$$|A| = \frac{1}{2.073} \approx 0.482$$

EXPECTED OUTPUT VOLTAGE

$$V_{out(peak)} = 0.482 \times 2 = 0.964V$$

$$V_{out(pp)} = 2 \times 0.964 = 1.93V$$

MEASURED OUTPUT VOLTAGE

From CRO observation: $V_{out(actual)} \approx 2V_{pp}$

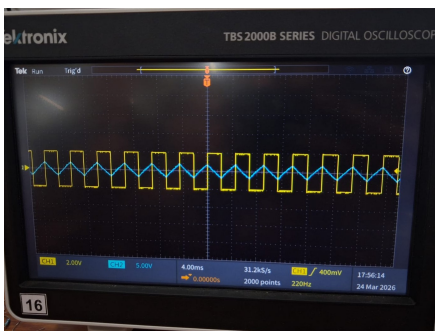
PERCENTAGE ERROR

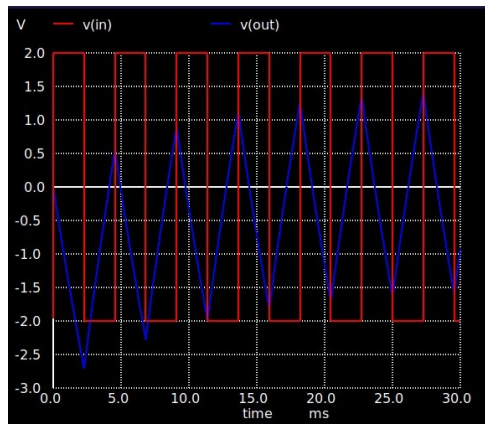
$$\text{Error} = \frac{V_{actual} - V_{ideal}}{V_{ideal}} \times 100 = \frac{2 - 1.93}{1.93} \times 100 \approx 3.6\%$$

ANALYSIS OF PRACTICAL DEVIATIONS

A slight amplitude mismatch and minor drift may be observed in CRO. This is due to op-amp offset voltage and finite gain-bandwidth. In simulation, ideal conditions eliminate these effects, resulting in a perfectly stable waveform.

0.2 For square wave input





For a square wave input, the signal remains constant over each half cycle. Since integration of a constant produces a linear variation, the output becomes a triangular waveform.

The amplitude is given by:

$$V_{out} = \frac{V_m}{RC} \cdot \frac{T}{2}$$

Theoretical Output:

$$V_{out} = \frac{V_{in}}{4fRC}$$

$$RC = 15000 \times 1 \times 10^{-7} = 1.5 \times 10^{-3}$$

$$4fRC = 4 \times 220 \times 1.5 \times 10^{-3} = 1.32$$

$$V_{out} = \frac{V_{in}}{1.32} = 0.758 V_{in}$$

For $V_{in} \approx 2V$:

$$V_{out(ideal)} = 0.758 \times 2 = 1.516 V$$

Measured Output: $V_{out(actual)} \approx 1.5 V$

Percentage Error:

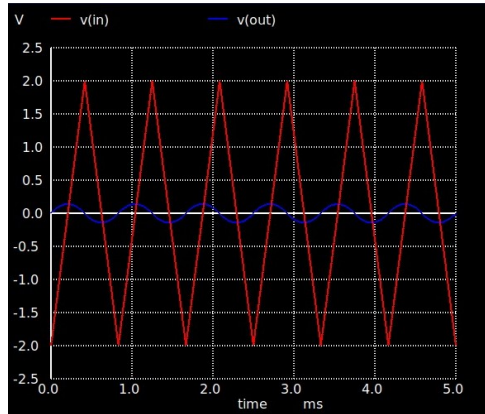
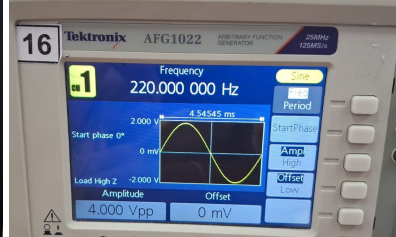
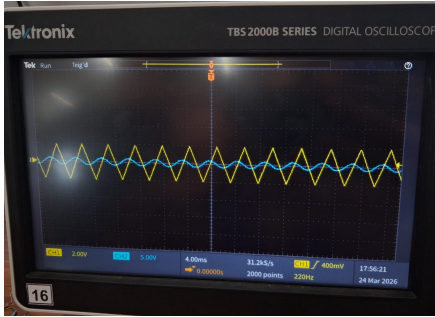
$$\begin{aligned} \text{Error} &= \frac{|V_{actual} - V_{ideal}|}{V_{ideal}} \times 100 \\ &= \frac{|1.5 - 1.516|}{1.516} \times 100 \approx 1.05\% \end{aligned}$$

Result: Voltage error $\approx 1\%$

Analysis of Practical Deviations The output triangular waveform may show slight tilting (drift) in CRO. This occurs due to accumulation of small DC offsets in the

integrator. In simulation, no DC offset exists, so the waveform remains perfectly symmetric.

0.3 For triangular wave input



Given: $f = 220 \text{ Hz}$, $R = 15 \text{ k}\Omega$, $C = 0.1 \mu\text{F}$

A triangular wave consists of linear segments. The integration of a linear function results in a parabolic waveform.

Thus, the output amplitude is given by:

$$V_{out} = \frac{V_m}{4fRC}$$

$$RC = (15 \times 10^3)(0.1 \times 10^{-6}) = 0.0015$$

$$\omega = 2\pi f = 2\pi(220) = 1382 \text{ rad/s}$$

Theoretical Gain:

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{\omega RC} = \frac{1}{1382 \times 0.0015} = 0.482$$

Observed Gain:

$$\left| \frac{V_{out}}{V_{in}} \right|_{\text{actual}} \approx 0.5$$

Error Calculation:

$$\% \text{ Error} = \frac{|0.5 - 0.482|}{0.482} \times 100 = 3.73\%$$

Waveform Analysis (From CRO Images)

- Yellow trace (CH1): Input signal
- Blue trace (CH2): Output signal
- The following transformations are verified:
 - Sine $\rightarrow$ Cosine (phase shift)
 - Square $\rightarrow$ Triangular
 - Triangular $\rightarrow$ Parabolic

Observation Table

Input Waveform	Input V_{pp}	Output V_{pp} (Ideal)	Output V_{pp} (Actual)	Error (%)
Sine Wave	4V	1.93V	2V	3.6%
Square Wave	4V	1.52V	1.5V	1.05%
Triangular Wave	4V	1.93V (via gain)	2V	3.73%

Conclusion

The op-amp integrator circuit was successfully implemented and tested. The observed output waveforms closely match theoretical predictions, confirming correct operation of the integrator with acceptable error.

Differentiator Circuit

Given Data

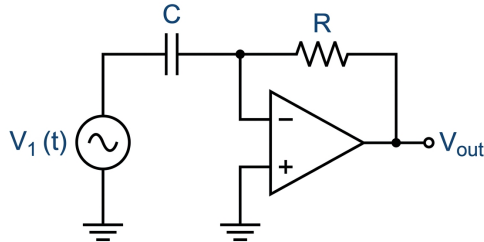
- Resistance, $R = 10 \text{ k}\Omega = 10^4 \Omega$
- Capacitance, $C = 100 \text{ nF} = 100 \times 10^{-9} \text{ F}$
- Input frequency, $f = 200 \text{ Hz}$

Theory

An op-amp differentiator produces an output proportional to the rate of change of the input.

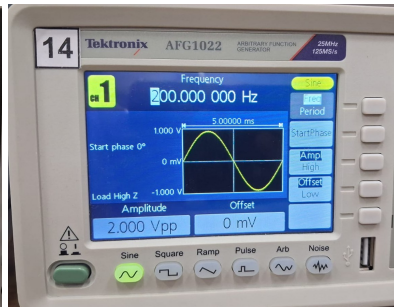
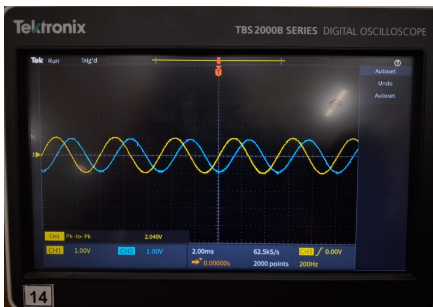
$$V_{out}(t) = -RC \frac{dV_{in}(t)}{dt}$$

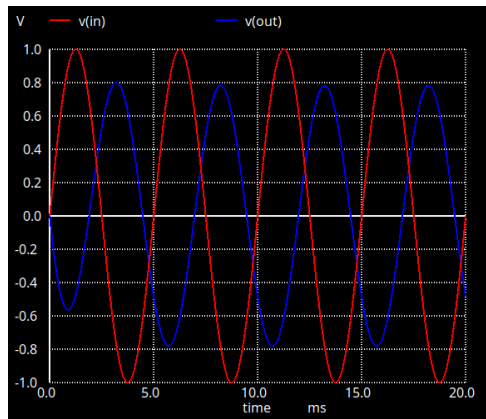
Circuit Diagram



Observation

- Input and output waveforms were observed using CRO.
- For sinusoidal input:
 - Output is sinusoidal with 90° phase shift.
 - Output amplitude is higher than input.





0.4 For sinusoidal input

$$|V_{out}| = \omega RC V_m, \quad \omega = 2\pi f$$

$$\omega = 2\pi \times 200 = 1256 \text{ rad/s} \quad RC = 10^4 \times 100 \times 10^{-9} = 0.001$$

$$|V_{out}| = 1256 \times 0.001 \times 1 = 1.256 \text{ V}$$

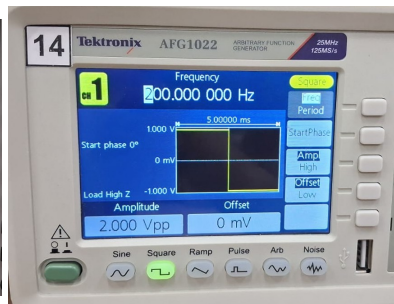
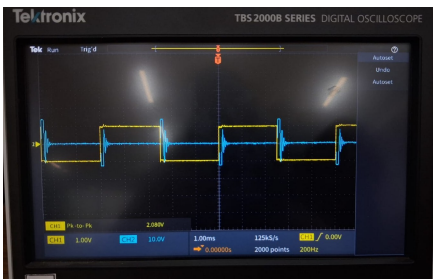
$$V_{out(pp)} = 2 \times 1.256 = 2.512 \text{ V}$$

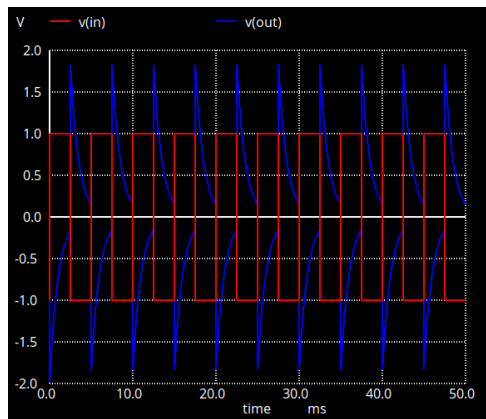
$$\text{Error: } V_{practical} \approx 2.4 V_{pp}$$

$$\% \text{ Error} = \frac{|2.4 - 2.512|}{2.512} \times 100 \approx 4.46\%$$

0.5 For square wave input

- Output shows sharp positive and negative spikes.
- Spikes occur at transitions of input.





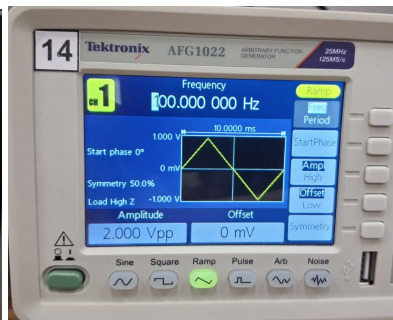
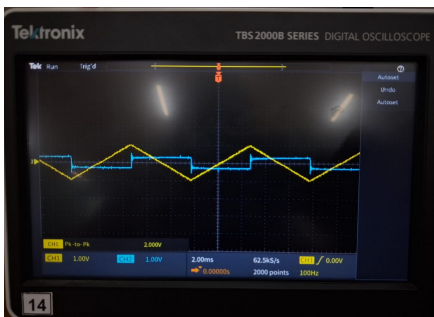
For a square wave input, the differentiator ideally produces sharp spikes at the rising and falling edges. This behavior was observed in simulation. In the practical setup, multiple spikes (2 to 3) were observed instead of a single spike due to noise, parasitic elements, and non-ideal characteristics of the op-amp.

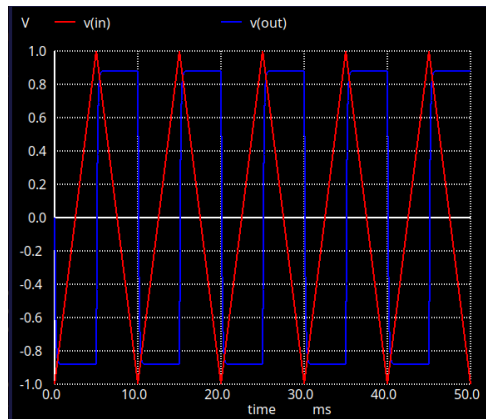
Note: Error calculation is not performed for square wave input because the ideal output consists of impulses (infinite amplitude and infinitesimally small duration) at the points of discontinuity.

In practical circuits, these impulses appear as finite spikes due to limitations of the op-amp bandwidth, slew rate, and circuit components. Hence, it is not possible to accurately compare theoretical and practical values.

0.6 For triangular wave input

- Output is a square waveform.





$$T = \frac{1}{f} = 0.01 \text{ s}$$

$$\frac{dV}{dt} = \frac{2}{T/2} = \frac{2}{0.005} = 400 \text{ V/s}$$

$$V_{out} = RC \cdot \frac{dV}{dt} = 0.001 \times 400 = 0.4 \text{ V} \quad V_{out(pp)} = 2 \times 0.4 = 0.8 \text{ V}$$

Error: $V_{practical(pp)} \approx 0.78 \text{ V}$

$$\% \text{ Error} = \frac{0.8 - 0.78}{0.8} \times 100 = 2.5\%$$

Waveform Analysis

- Yellow trace (CH1): Input signal
- Blue trace (CH2): Output signal
- Frequency observed: $\approx 200 \text{ Hz}$
- Transformations verified:
 - Sine $\rightarrow$ Cosine
 - Square $\rightarrow$ Spikes
 - Triangular $\rightarrow$ Square

Observation Table (Values)

Input Type	$V_{in(pp)}$	$V_{out(pp)}$ (Theoretical)	$V_{out(pp)}$ (Practical)	% Error
Sine	2V	2.512V	2.4V	4.46%
Square	2V	Ideal (Impulse)	Finite spikes	–
Triangle	2V	0.8V	0.78V	2.5%

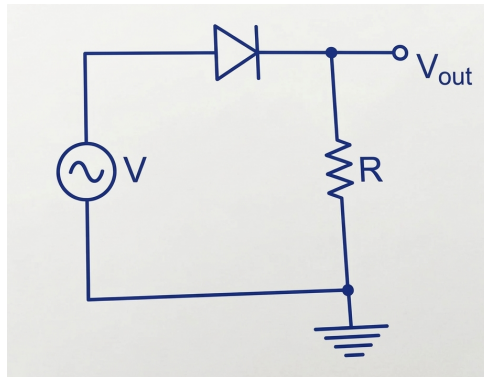
Half-Wave Rectifier (Without Op-Amp)

Theory

A half-wave rectifier using a diode allows only one half of the input signal to pass while blocking the other half.

$$V_{out} = \begin{cases} V_{in} - V_D, & V_{in} > V_D \\ 0, & V_{in} < V_D \end{cases}$$

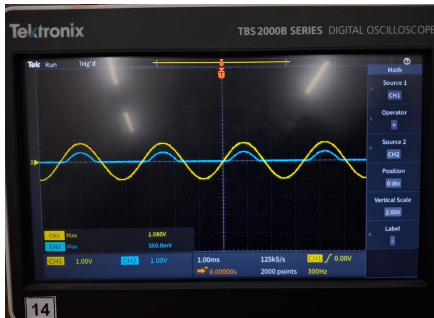
where $V_D \approx 0.7V$ is the diode forward voltage.



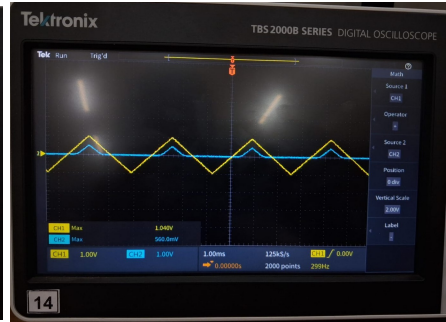
Working

- During positive half cycle:
 - Diode is forward biased
 - Output follows input minus diode drop
- During negative half cycle:
 - Diode is reverse biased
 - Output is zero

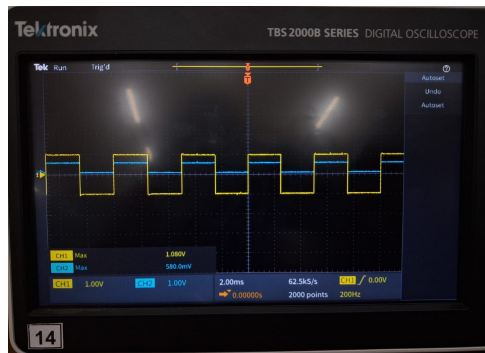
Observations



sine wave input



Triangular wave input



square wave input

The reduction in output amplitude and distortion near zero crossing are due to the diode forward voltage drop.

Observations

- Only positive half cycles appear at the output
- Output amplitude is reduced due to diode forward voltage drop
- Small signals near zero are not rectified

The limitations of the conventional half-wave rectifier are overcome by using a precision rectifier.

Half-Wave Precision Rectifier

A half-wave precision rectifier uses an op-amp and diode to eliminate the diode threshold error (approximately 0.7V).

$$V_{out} = \begin{cases} V_{in}, & V_{in} > 0 \\ 0, & V_{in} < 0 \end{cases}$$

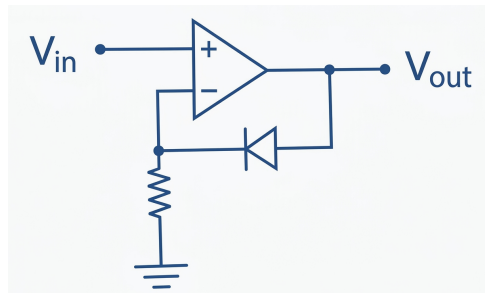
Output behavior:

- For positive input → output follows input
- For negative input → output is approximately zero

Advantage:

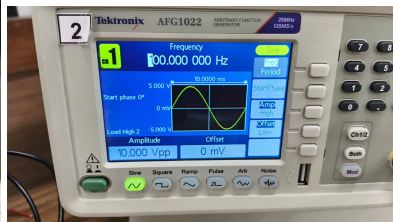
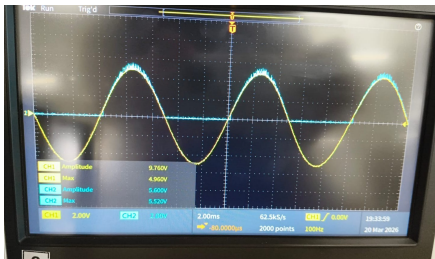
- Can rectify small signals accurately compared to a normal diode rectifier

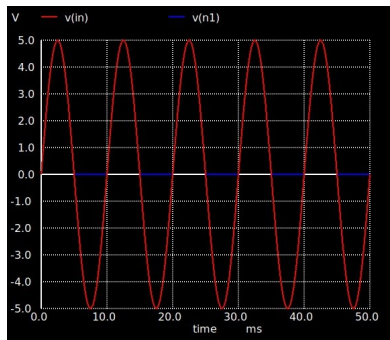
Circuit Diagram



Observations

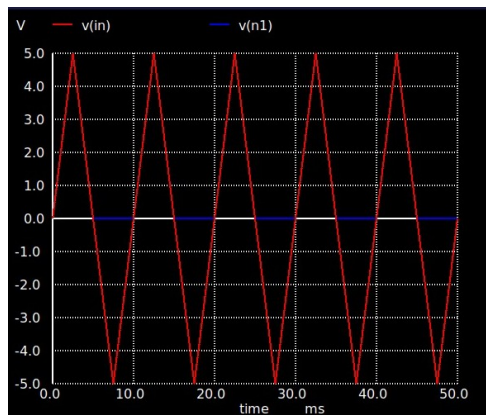
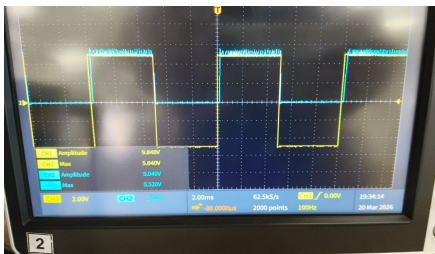
(A) Sine Wave Input





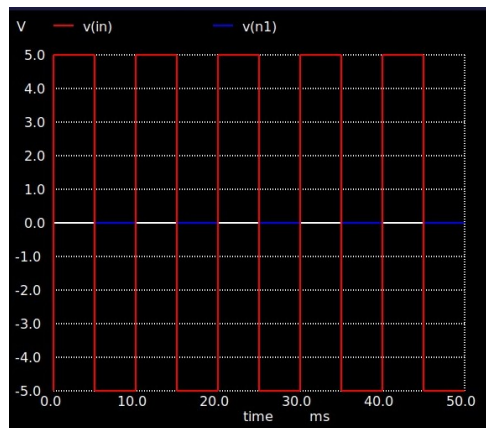
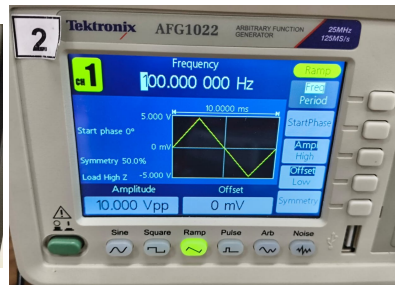
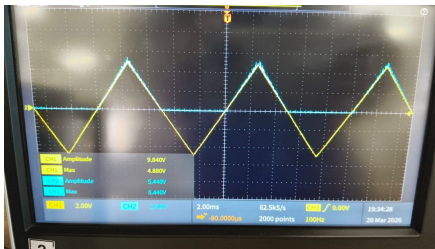
Observation: The output follows the positive half cycle of the input sine wave and suppresses the negative half cycle, confirming rectification.

(B) Square Wave Input



Observation: The output becomes a unipolar square wave, where negative levels are clipped to zero.

(C) Triangular Wave Input



Observation: Only the positive triangular portions are observed at the output, while negative portions are clipped.

Observation Table

Input Type	Input Shape	Output Shape	Observation
Sine	Sinusoidal	Half-wave	Negative clipped
Square	Square	Pulses	Only positive level
Triangle	Triangle	Half triangle	Negative removed

Waveform Explanation

- The op-amp amplifies the input before the diode
- This removes the diode threshold effect
- During positive cycle:

- Diode conducts
 - Output follows input
- During negative cycle:
 - Diode is OFF
 - Output is approximately zero

Conclusion

The half-wave precision rectifier was successfully implemented and tested. The circuit accurately rectifies input signals by eliminating the diode threshold voltage, producing output only during positive half cycles for all input waveforms.